

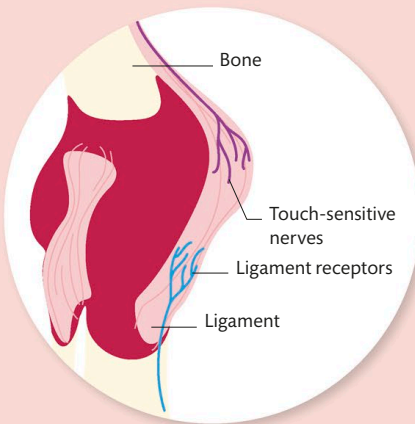


GROWTH SPURTS CAN CONFUSE THE BRAIN AS IT CANNOT KEEP UP WITH CHANGES IN LIMB DIMENSIONS



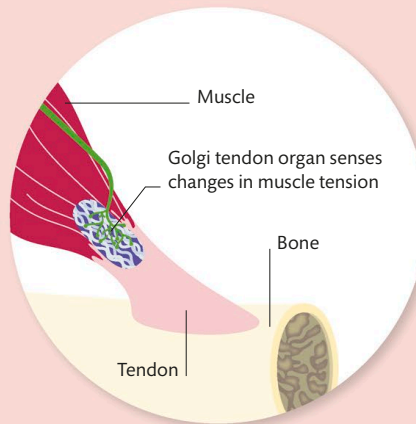
Types of proprioceptors

The body contains a variety of proprioceptors, and the combined information from these receptors helps the brain construct an overall picture of the body's position. There are three main types of proprioceptors: muscle spindle fibers, which are embedded in our muscles; Golgi tendon organs, which are located at the junction between tendons and muscles; and joint receptors, which attach to our joints. Special receptors in the skin can also detect stretch (see p.83).



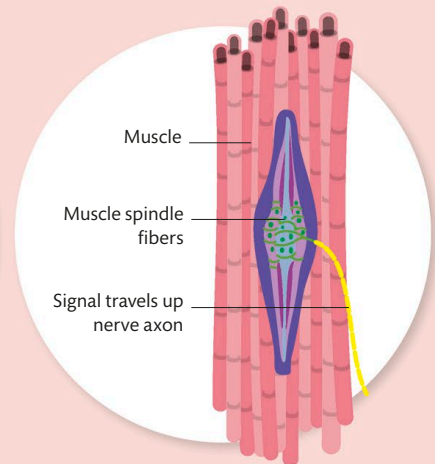
Joint receptors

Nerve endings within our joints detect the joints' position. The receptors help prevent damage through overextension as well as detecting position in normal motion.



Tendon receptors

Golgi tendon organs are found within the tendons at the ends of muscles. They monitor muscle tension to ensure we do not overstretch the muscles.

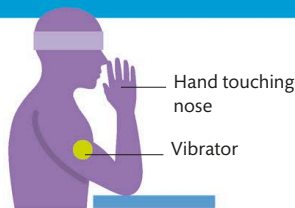


Muscle receptors

Muscles have position sensors called spindle fibers within them. As they stretch, the spindles send information to the brain about the positions of the muscles.

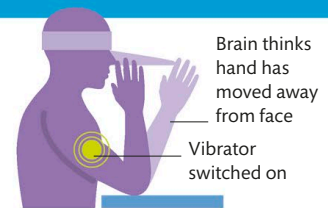
THE PINOCCHIO ILLUSION

Sometimes proprioception can be confused, making the body feel like something is happening when it is not. One such effect is called the Pinocchio illusion. A vibrator is fixed to a person's bicep. If the person holds her nose while the vibrator is turned on, she will feel as though her arm is moving away from her nose. It happens because the vibrator stimulates the muscle spindle fibers in the biceps in the same way as if the muscle was stretching. Because the fingers are still touching the nose, it feels as if the nose is growing out from the face.



Before stimulation

At rest, the brain is aware that the fingers are touching the nose, but there is no movement of the arm.



During stimulation

Vibrations tell the brain that the arm is moving, creating a sensation that the nose is growing outward.